Automotive Sensors

Introduction of Kalman Filter

Automotive Intelligence Lab.

Contents

Bayes filter

Bayes' Theorem

- **Thomas Bayes (1701-1761)**

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}$$

Bayes' Theorem (I)

- **We can obtain a conditional probability of a hypothesis given new evidence.**
  - Hypothesis, x :a proposed explanation for a phenomenon
  - Evidence, z : information indicating whether a belief or proposition is true or valid

Probability of the Evidence given the Hypothesis

The Prior Probability of the Hypothesis

The Posterior probability of the Hypothesis given the Evidence

$$P(x|z) = \frac{P(z|x)\ P(x)}{P(z)}$$

The Prior Probability of the Evidence

Bayes' Theorem (II)

- **Law of total probability (marginalization)**
  - The prior probability of the evidence can be represented as

$$P(z) = \sum_{x'} P(z|x')\ P(x')$$

The Posterior probability of the Hypothesis given the Evidence

Probability of the Evidence given the Hypothesis

The Prior Probability of the Hypothesis

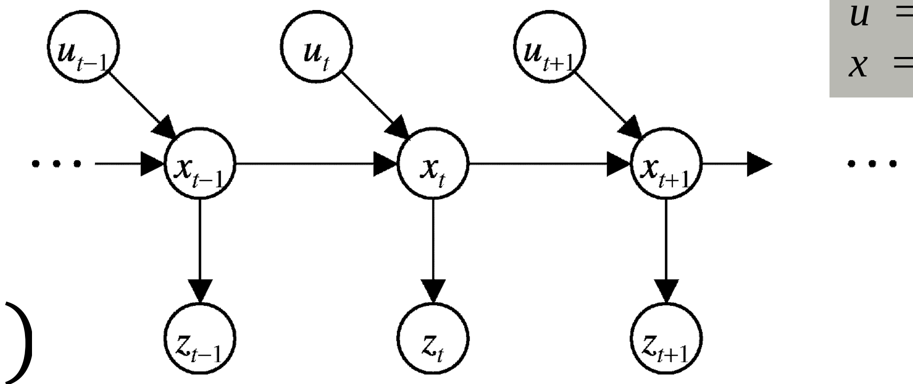
$$P(x|z) = \frac{P(z|x)\ P(x)}{\sum_{x'} P(z|x')\ P(x')} = \mu P(z|x) P(x)$$

Normalization factor

Markov Assumption

- **The current state is conditionally independent of all the other past states in the previous time stamp.**
- **The evidence z_t at time t depends only on the state at time t.**
- **The Past and future data are independent if one knows the current state x_t.**
- **Underlying Assumptions**
  - Static world
  - Independent noise
  - Perfect model, no approximation errors

z = observation
u = action
x = state



$$p(z_t | x_{0:t}, z_{1:t}, u_{1:t}) = p(z_t | x_t)$$

$$p(x_t | x_{1:t-1}, z_{1:t}, u_{1:t}) = p(x_t | x_{t-1}, u_t)$$

Bayes Filter

z = observation
u = action
x = state

$$\boxed{Bel(x_t)} = P(x_t|u_{1:t}, z_{1:t})$$

Bayes $$= \eta\ P(z_t|x_t, u_{1:t}, z_{1:t-1})\ P(x_t|u_{1:t}, z_{1:t-1})$$

Markov $$= \eta\ P(z_t|x_t)\ P(x_t|u_{1:t}, z_{1:t-1})$$

Total prob. $$= \eta\ P(z_t|x_t) \int P(x_t|x_{t-1}, u_{1:t}, z_{1:t-1})\, P(x_{t-1}|u_{1:t}, z_{1:t-1})\ dx_{t-1}$$

Markov $$= \eta\ P(z_t|x_t) \int P(x_t|x_{t-1}, u_t)\ P(x_{t-1}|u_{1:t}, z_{1:t-1})\ dx_{t-1}$$

Markov $$= \eta\ P(z_t|x_t) \int P(x_t|u_t, x_{t-1})\, P(x_{t-1}|u_{1:t-1}, z_{1:t-1})\ dx_{t-1}$$

$$\boxed{= \eta\ P(z_t|x_t) \int P(x_t|u_t, x_{t-1})\ Bel(x_{t-1})\ dx_{t-1}}$$

Bayes Filter Algorithm

$$Bel(x_t) = \eta\ P(z_t|x_t) \int P(x_t|u_t, x_{t-1})\ Bel(x_{t-1})\ dx_{t-1}$$

1. **Algorithm Bayes_filter($\boldsymbol{Bel(x_t), d}$):**
2. $\boldsymbol{\alpha}$ **= 0**
3. **If *d* is a perceptual data item z_t then**
4. **For all x_t do**
5. $Bel'(x_t) = P(z_t|x_t)Bel(x_t)$
6. $\alpha = \alpha + Bel'(x_t)$
7. **For all x_t do**
8. $Bel'(x_t) = \alpha^{-1}Bel'(x_t) = \eta Bel'(x_t)$
9. **Else if *d* is an action data item u_t then**
10. **For all x_t do**
11. $Bel'(x_t) = \int P(x|u_t, x'_{t-1})\ Bel(x'_{t-1})\ dx'_{t-1}$
12. **Return $\boldsymbol{Bel'(x_t)}$**

Bayes Filters are Familiar!

$$Bel(x_t) = \eta \ P(z_t|x_t) \ \int P(x_t|u_t, x_{t-1}) \ Bel(x_{t-1}) \ dx_{t-1}$$

- Kalman filters
- Particle filters
- Hidden Markov models
- Dynamic Bayesian networks
- Partially Observable Markov Decision Processes (POMDPs)

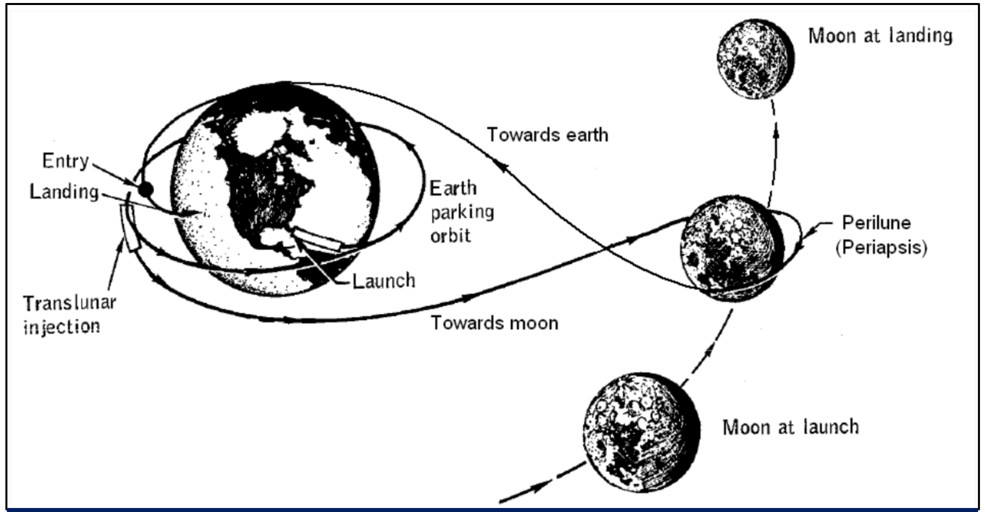
Kalman filter

History of Kalam Filter

- Developed by Rudolf E. Kálmán in 1960
- Operates by combining 2 methods: Prediction & measurement
- Successfully used in the Apollo navigation system

Rudolf E. Kálmán receiving the National Medal of Science

Apollo Guidance Computer



The (extended) Kalman Filter became widely known after its use in the Apollo Guidance Computer for circumlunar navigation.

Kalman filter?

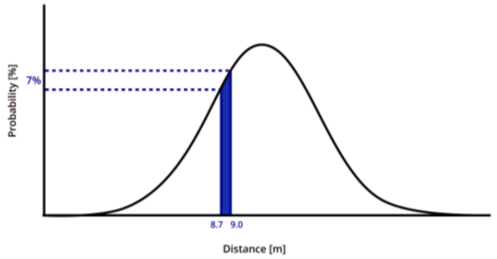
- The type of Bayes filter using "Linear" system and "Gaussian" probability distribution

$$Bel(x_t) = \eta \ P(z_t|x_t) \int P(x_t|u_t, x_{t-1}) \ Bel(x_{t-1}) \ dx_{t-1}$$

$$x[k+1] = Ax[k] + Bu[k]$$
$$y[k] = Cx[k] + Du[k]$$

Linear system

$$p(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$



Gaussian distribution

What is uncertainty?

Kalman Filter

Two-step Process

Use Information We Have to Predict the State

Predict State

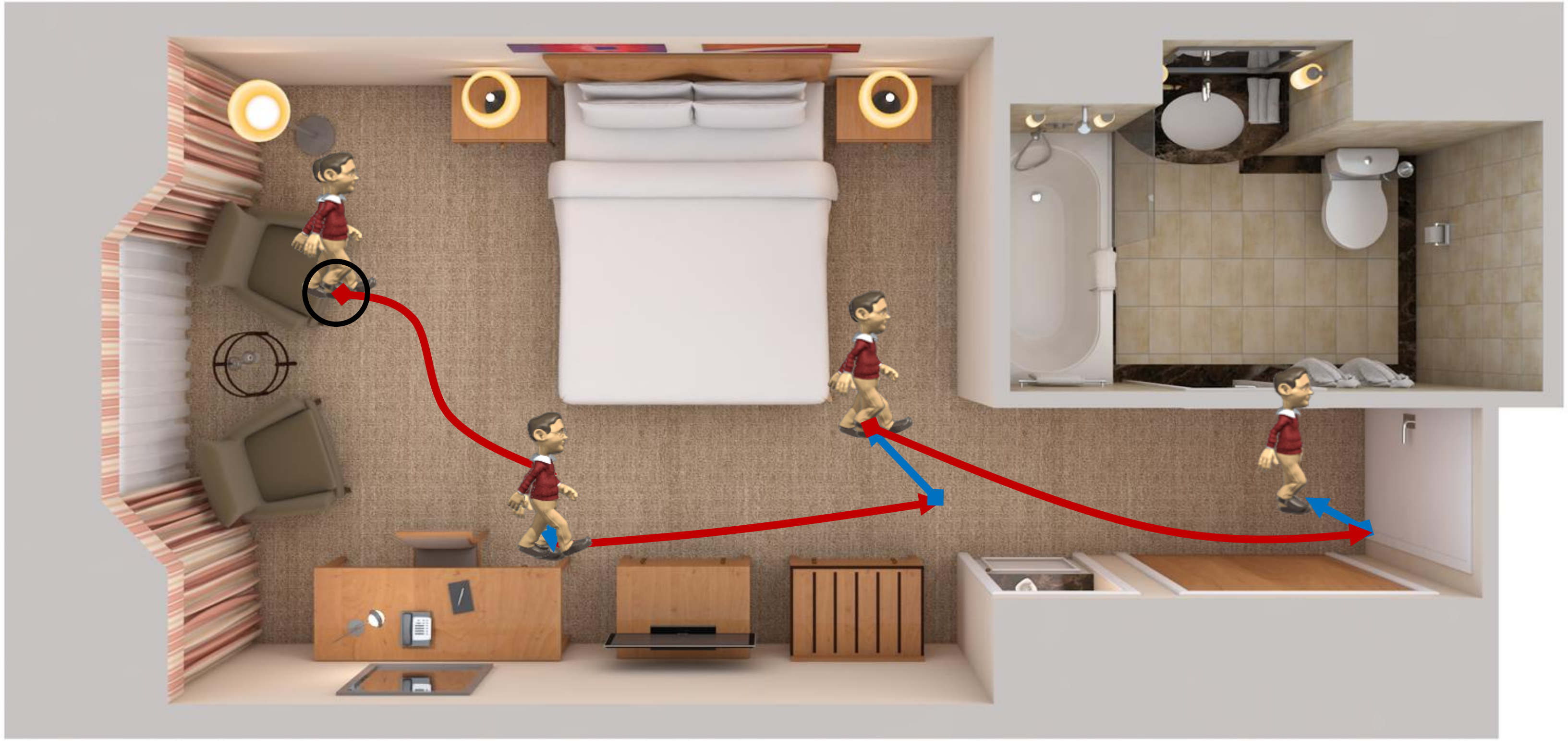
Update Measurement

Use New Observations to Correct Our Belief

Basic concept of Bayesian filter localization

- **Iterative Prediction (dead-reckoning) & Update (correction)**
  - Conceptual study: :10 step ➔ : stop

Initialization → **Prediction** → **Update** → **Prediction** → **Update** → **Prediction** → **Update**



Update Measurement

Two-step Process

Use Information We Have to Predict the State

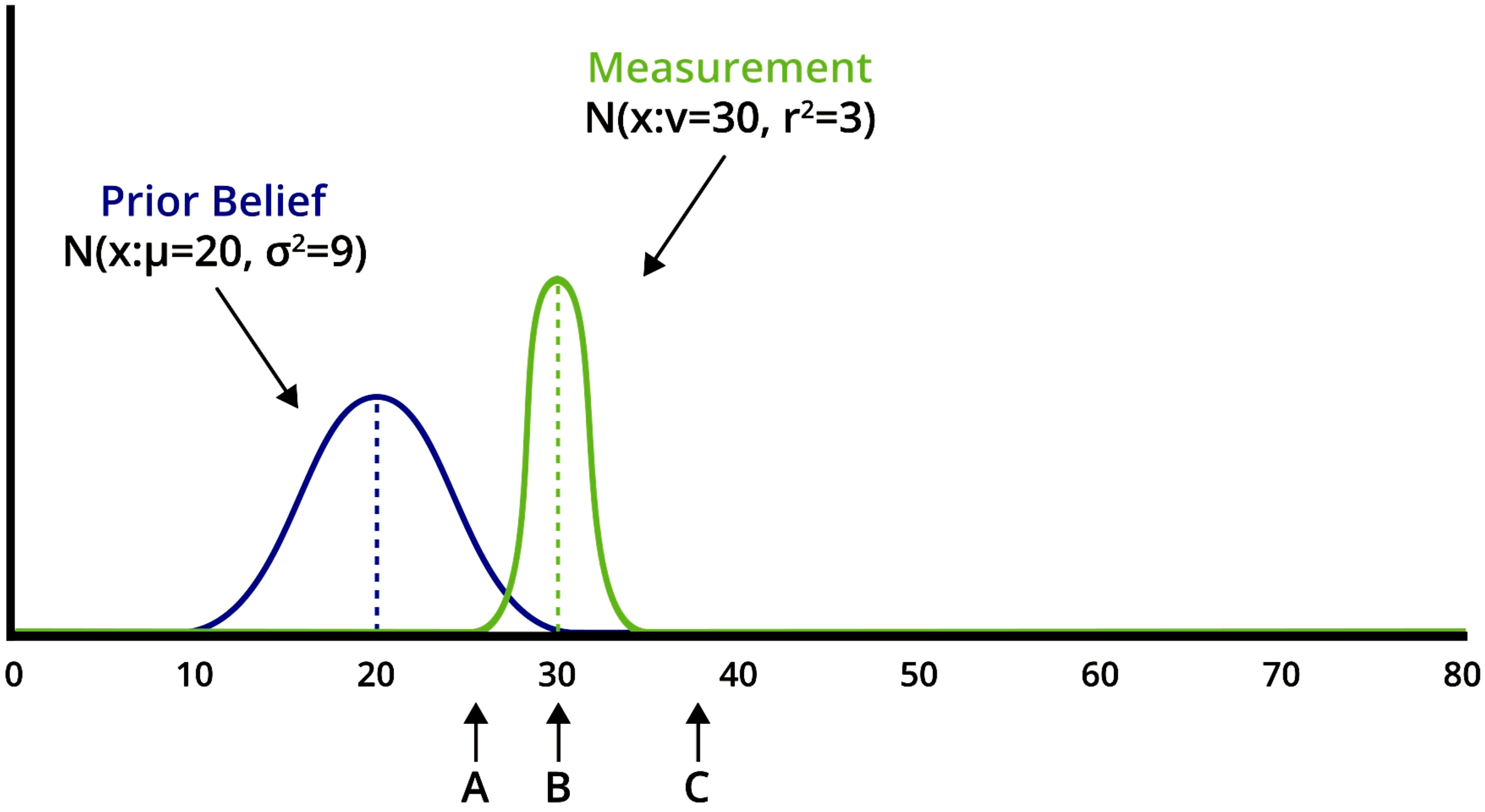
Predict State

Update Measurement

Use New Observations to Correct Our Belief

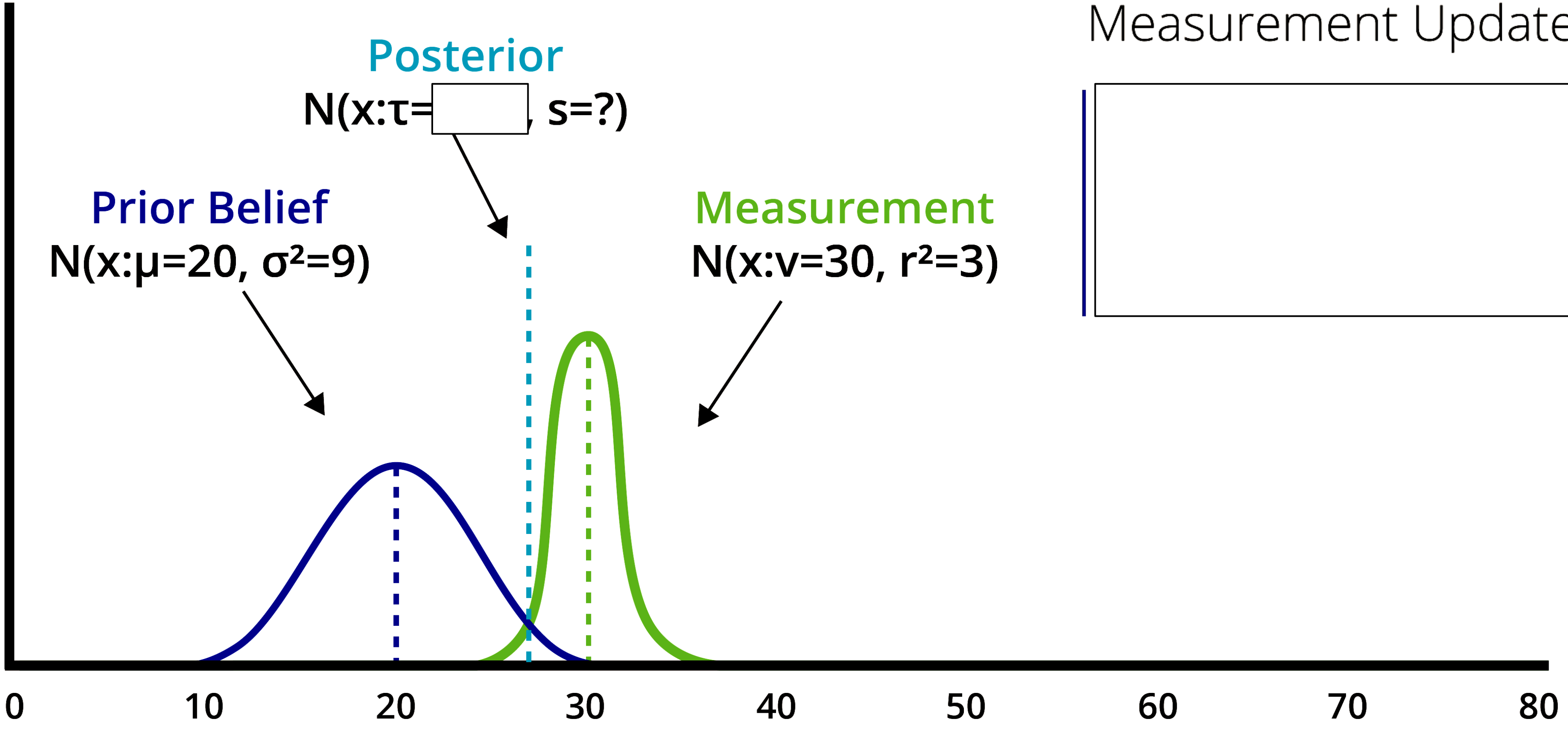
1D Kalman Filters – Measurement Update

■ Before measurement update



1D Kalman Filters – Measurement Update

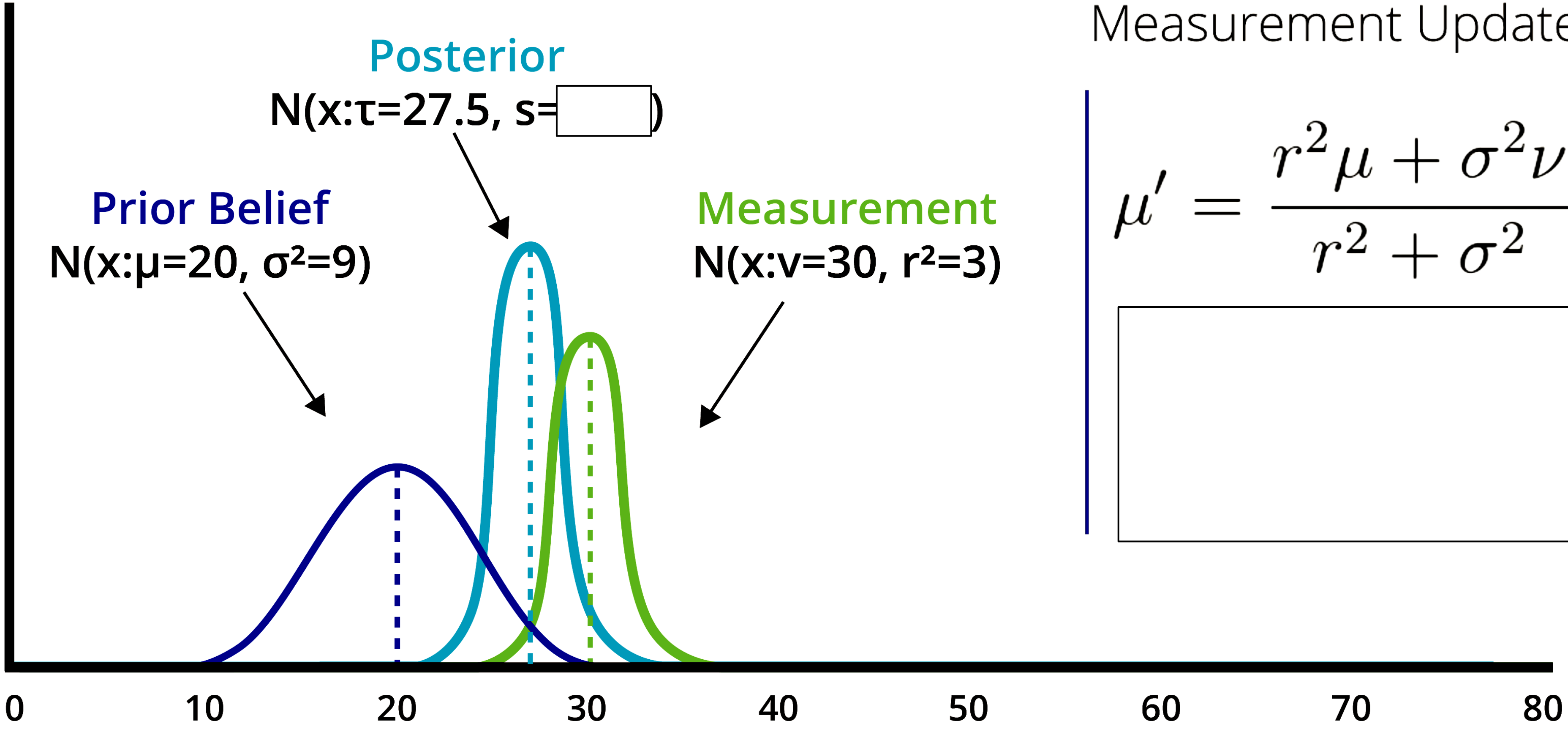
■ **After measurement update**



Measurement Update

1D Kalman Filters – Measurement Update

- After measurement update

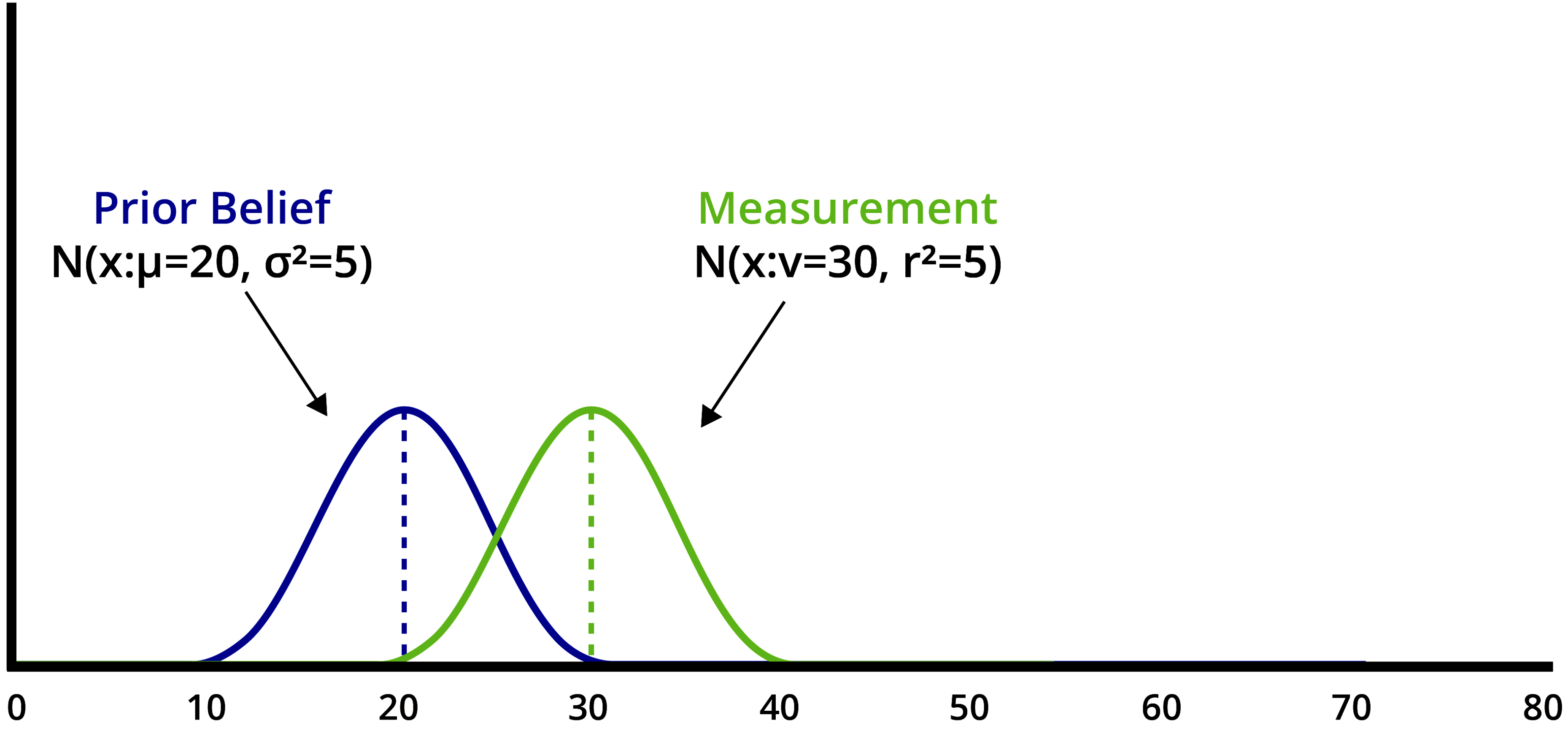


Measurement Update

$$\mu' = \frac{r^2\mu + \sigma^2\nu}{r^2 + \sigma^2}$$

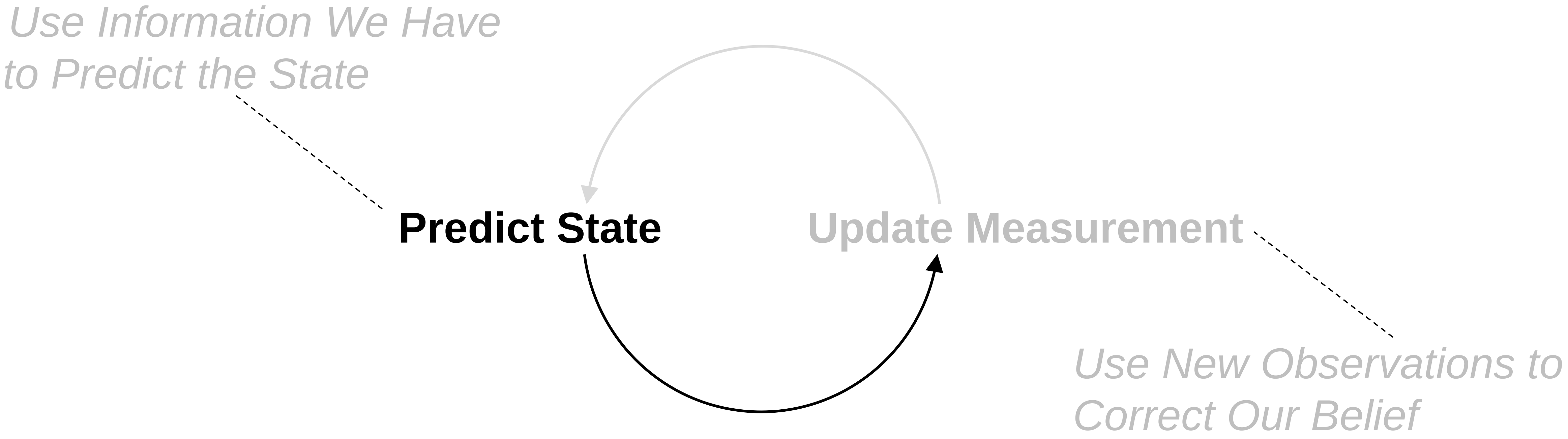
1D Kalman Filters – Measurement Update

Question?

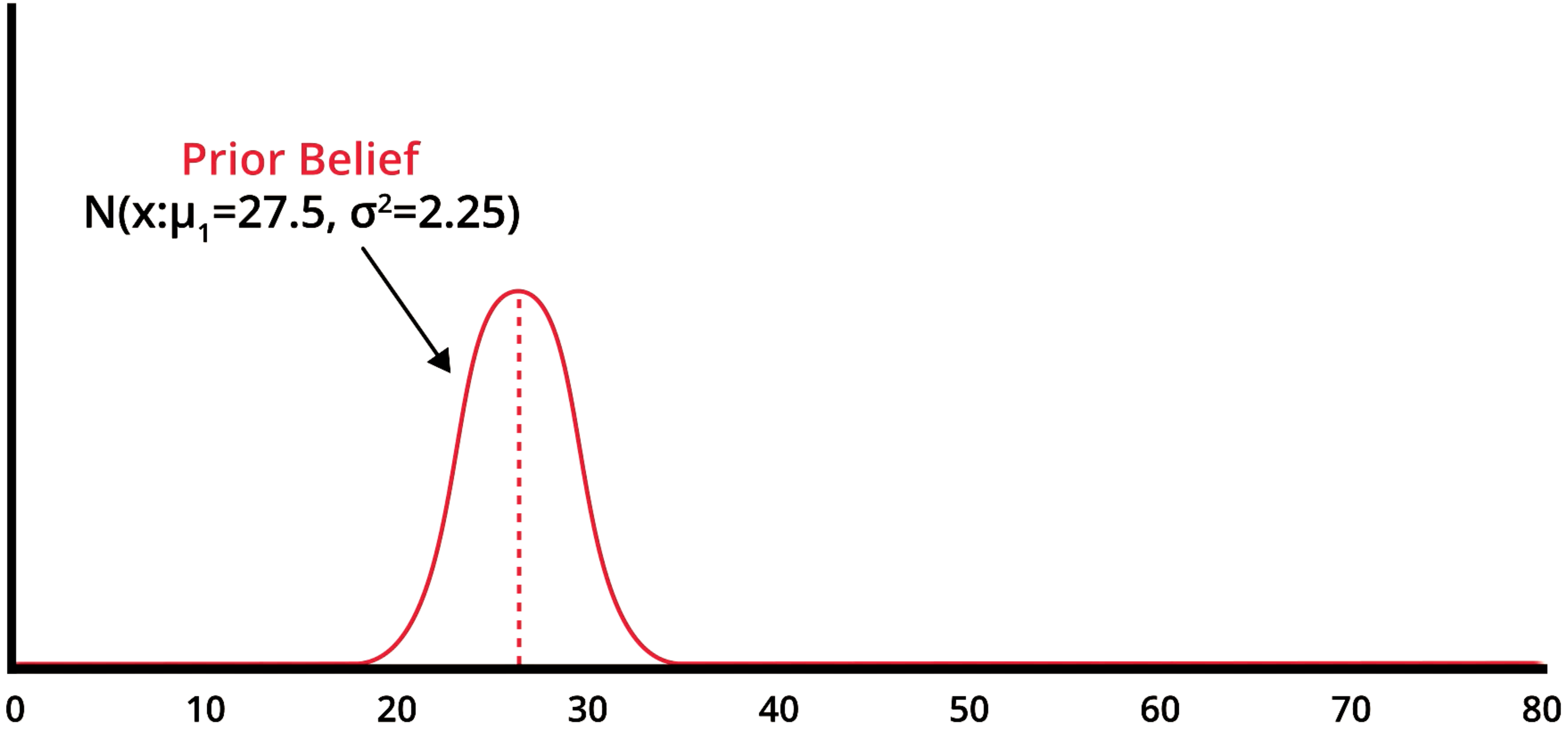


Predict State

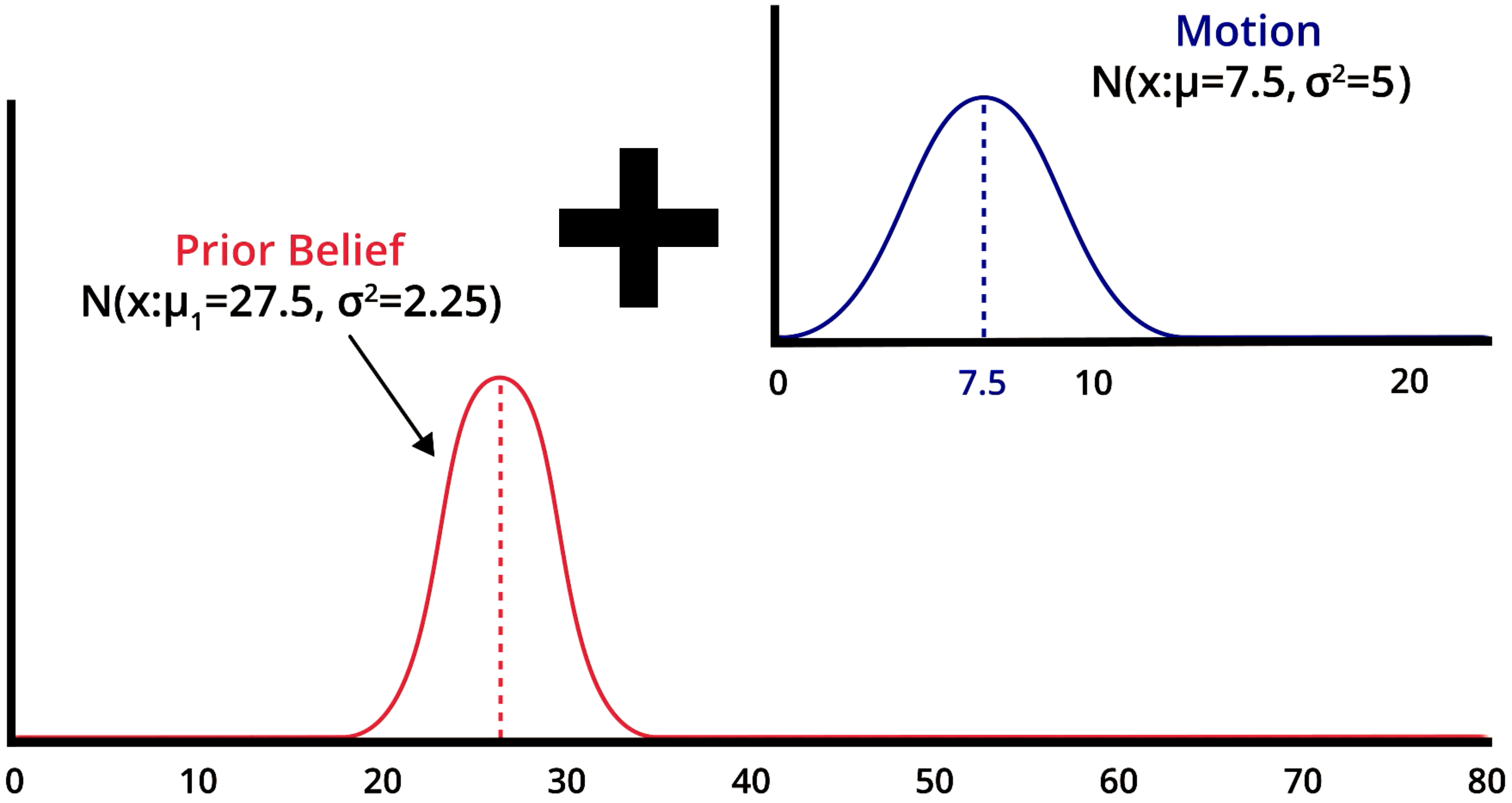
Two-step Process



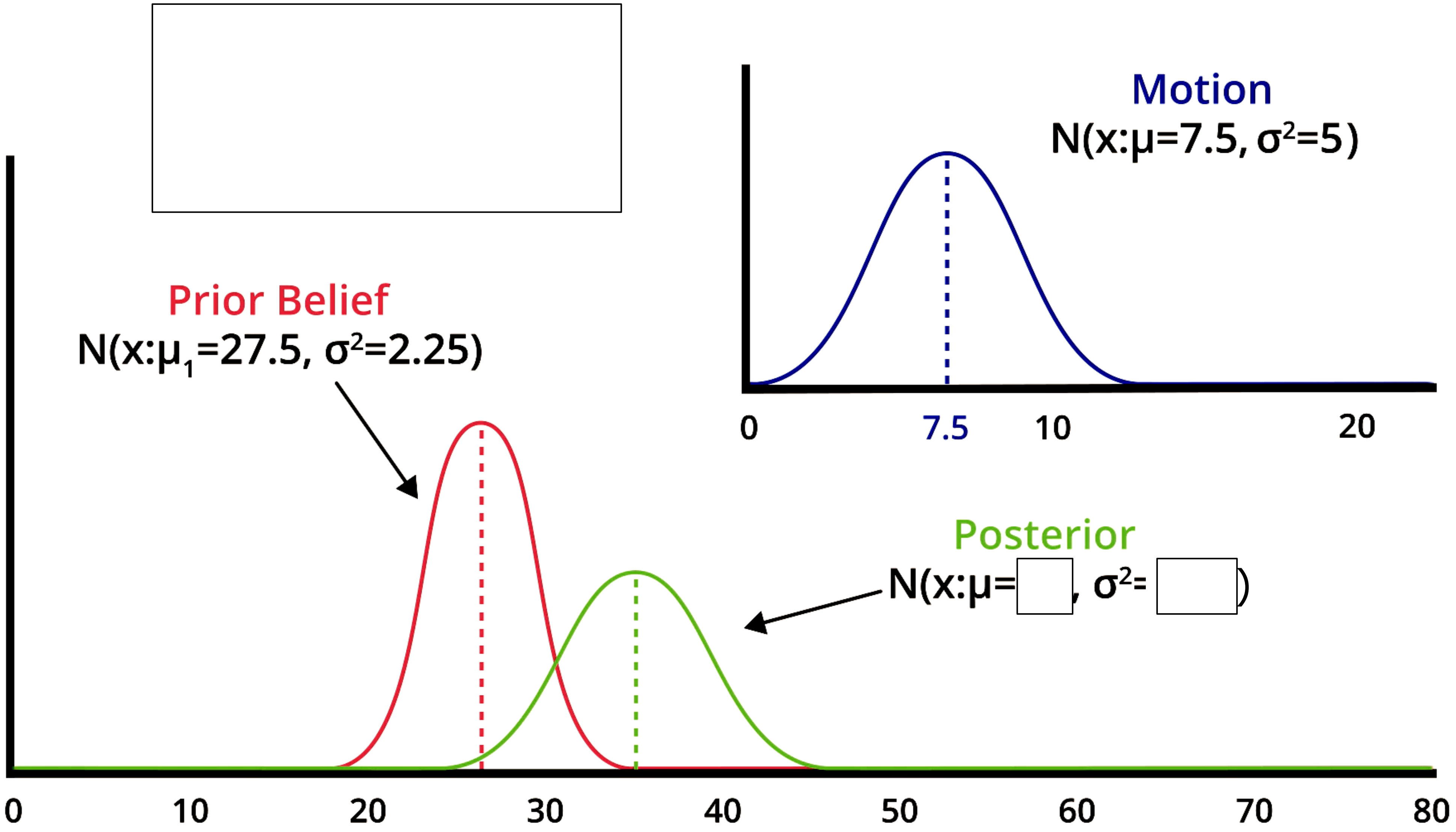
1D Kalman Filters – State Prediction



1D Kalman Filters – State Prediction



1D Kalman Filters – State Prediction



1D Kalman Filters

State Prediction

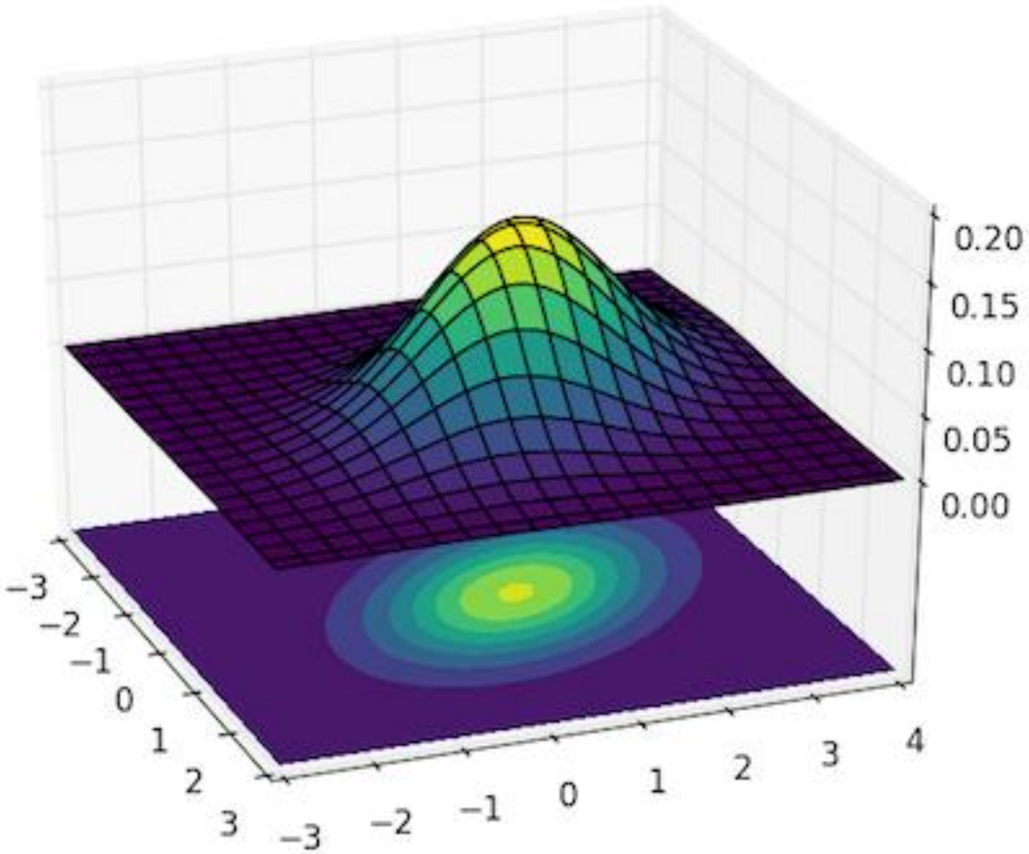
Measurement Update

Multivariate Gaussians

- **Most system that we would be interested in modeling are moving in more than one dimension.**
  - For instance, a vehicle on a plane would have an x & y position.

- **The simple approach to take, would be to have a 1-dimensional Gaussian represent each dimension - one for the x-axis and one for the y-axis.**
  - What's wrong with this?
  - There may be correlation between dimensions that we would net be able to model by using independent 1-dimensional Gaussians.

Multivariate Gaussians



Mean

- The mean is now a vector,

$$\mu = \boxed{}$$

Covariance

- The variance is a covariance matrix,

$$\Sigma = \boxed{}$$

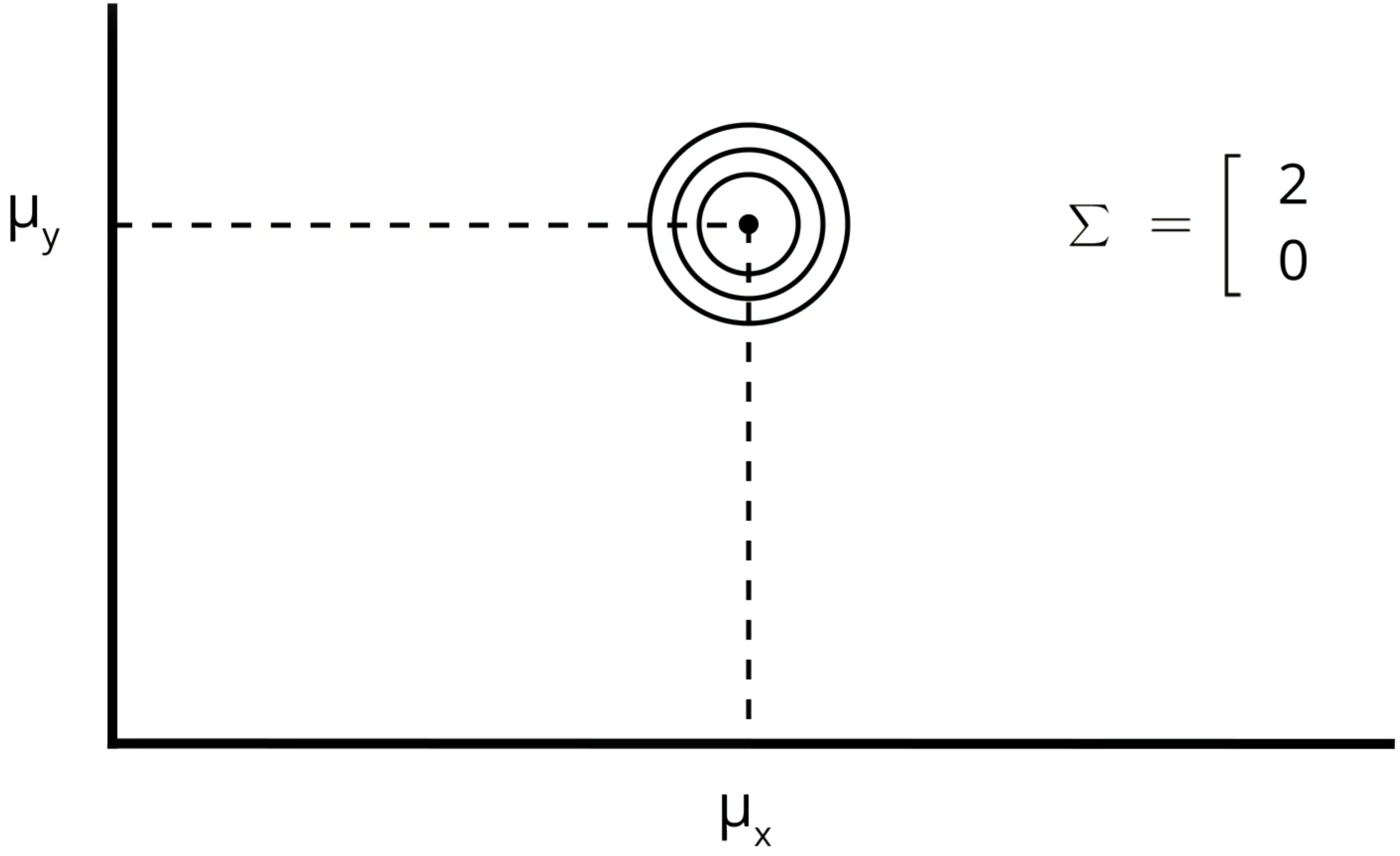
Multivariate Gaussian

- If D=1, the formula simplifies to the formula for the one-dimensional Gaussian that you have seen before.

$$p(x) = \boxed{}$$

Multivariate Gaussians

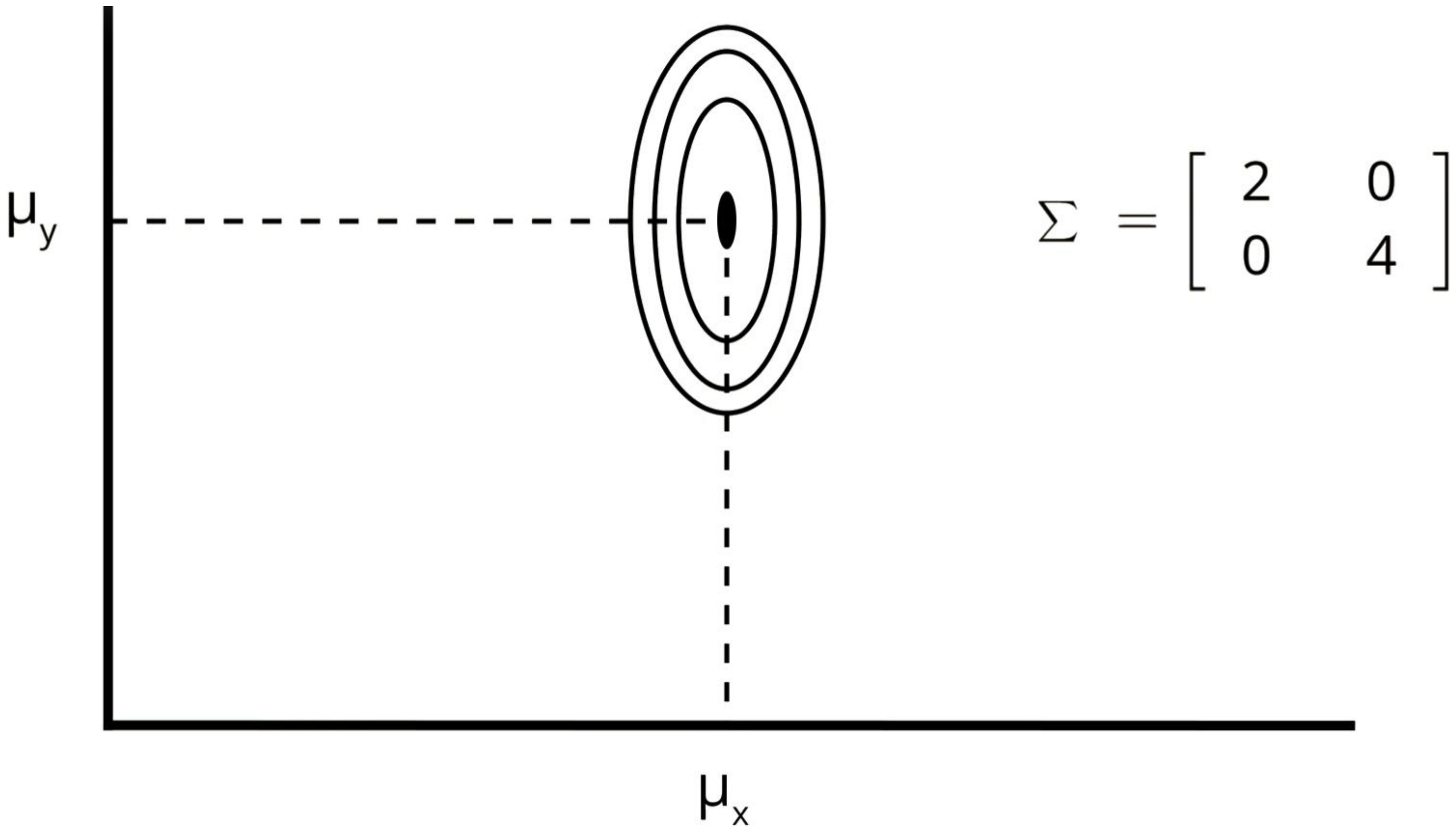
- Example of covariance



$$\Sigma = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

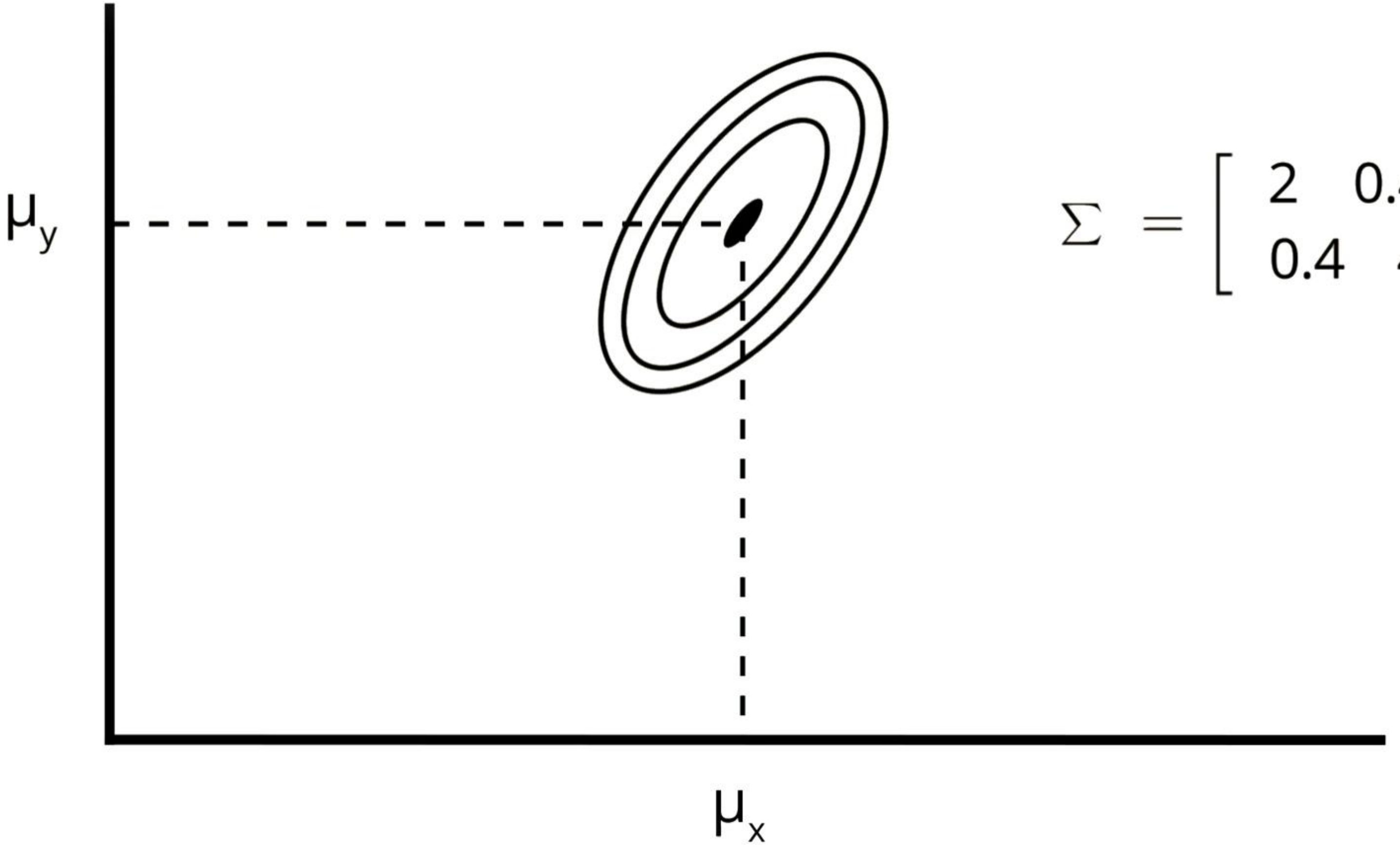
Multivariate Gaussians

Example of covariance



Multivariate Gaussians

Example of covariance



$$\Sigma = \begin{bmatrix} 2 & 0.4 \\ 0.4 & 4 \end{bmatrix}$$

Intro to Multidimensional KF

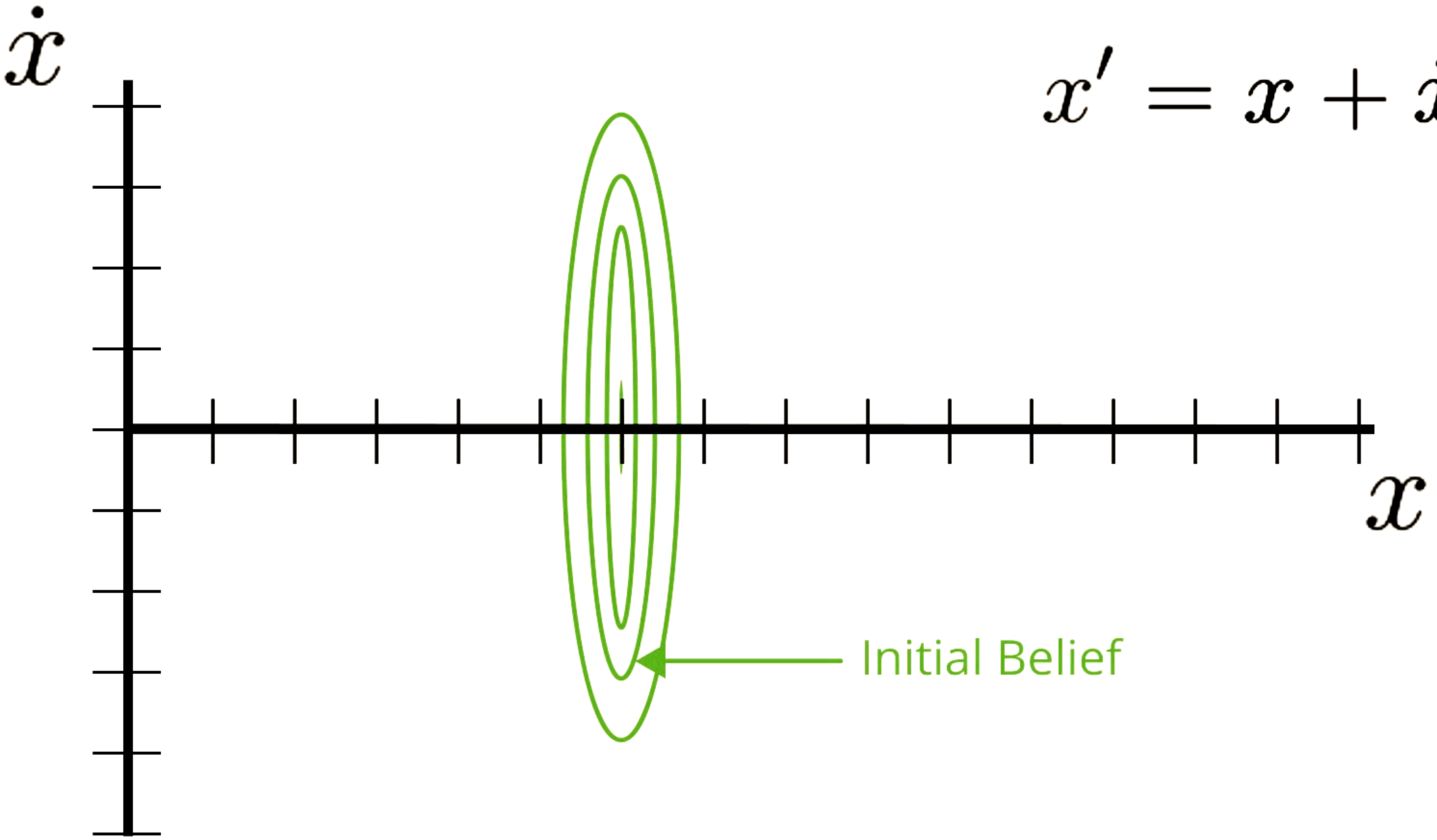
$$\begin{array}{r} \\ \text{Hidden} \longrightarrow \end{array} \begin{bmatrix} x \\ \dot{x} \end{bmatrix} \begin{array}{l} \longleftarrow \text{Observable} \\ \\ \end{array}$$

Intro to Multidimensional KF

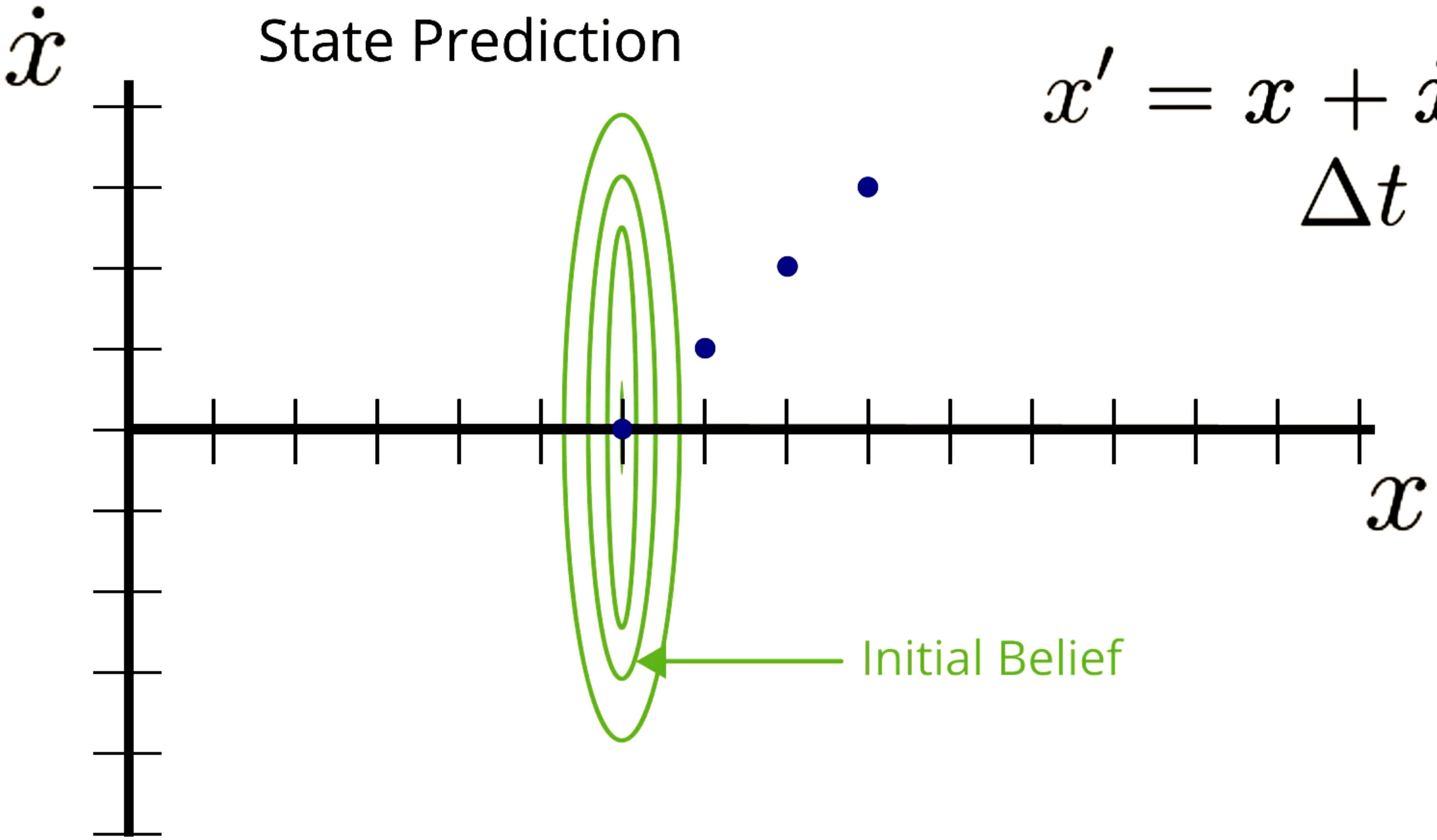
$$x' = x + \dot{x}\Delta t$$

Multidimensional Kalman Filters – Initial Estimate

$$x' = x + \dot{x}\Delta t$$

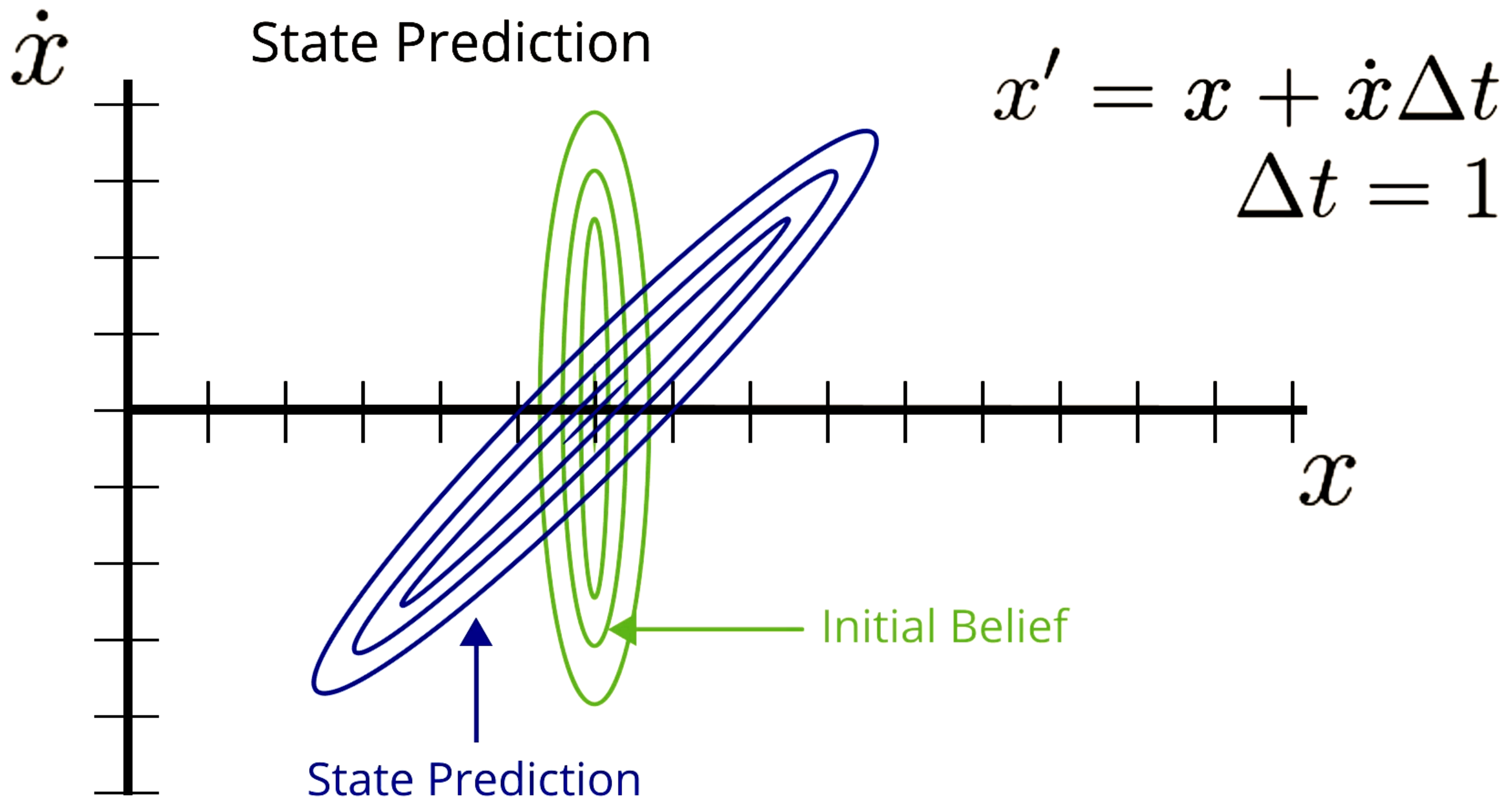


Multidimensional Kalman Filters – State Prediction



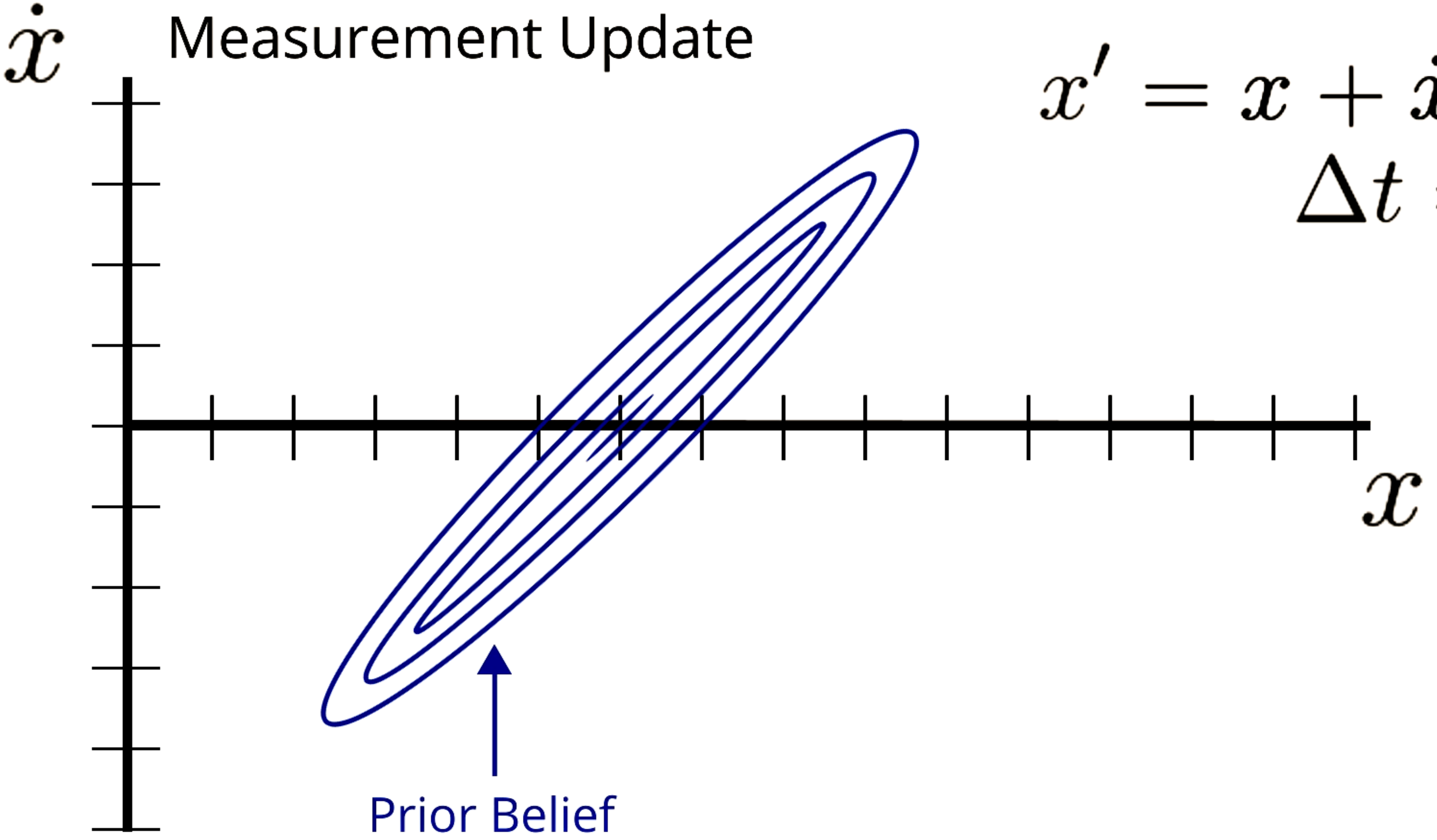
$$x' = x + \dot{x}\Delta t$$
$$\Delta t = 1$$

Multidimensional Kalman Filters – State Prediction



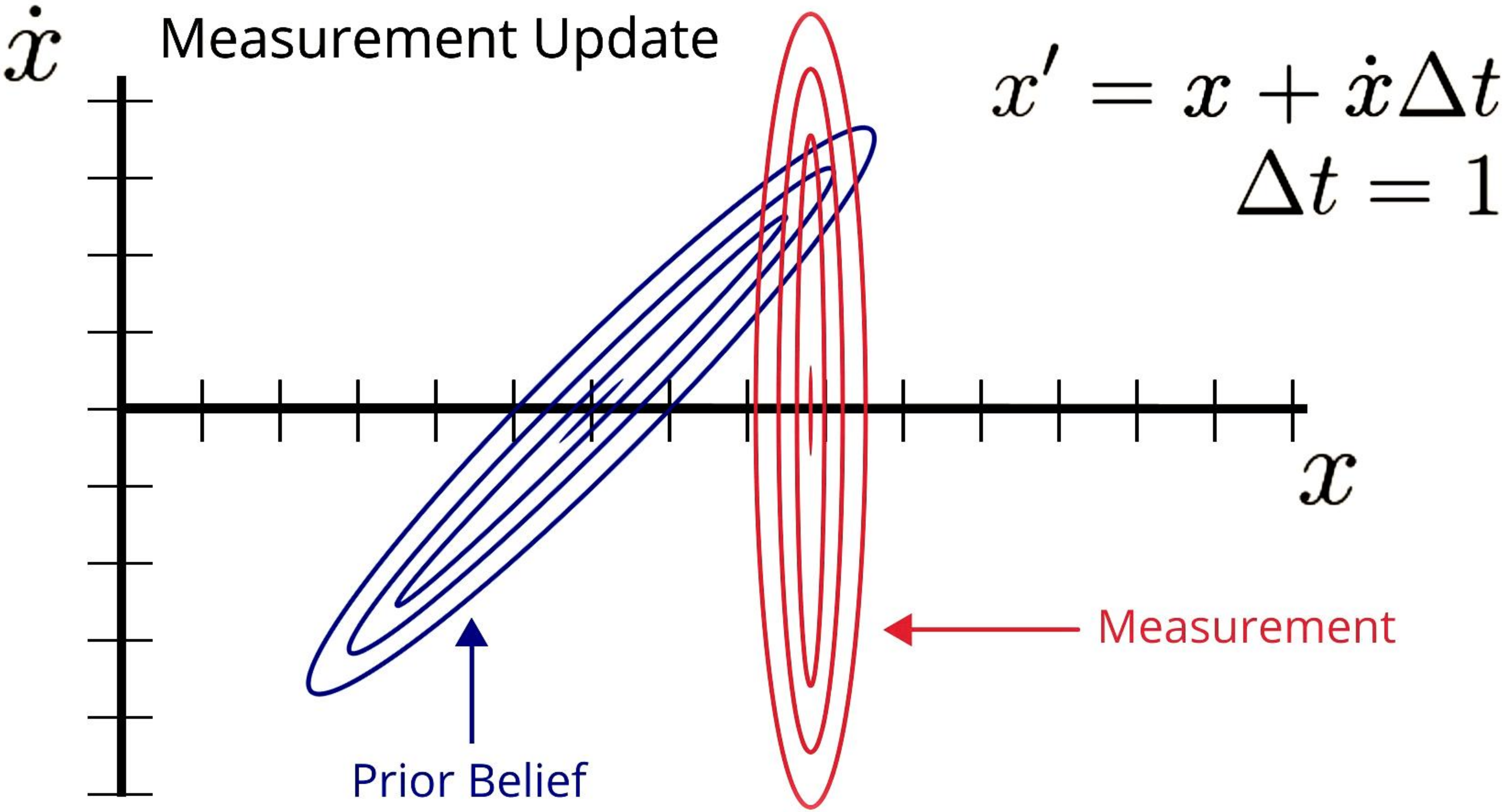
Multidimensional Kalman Filters – Measurement Update

Measurement Update

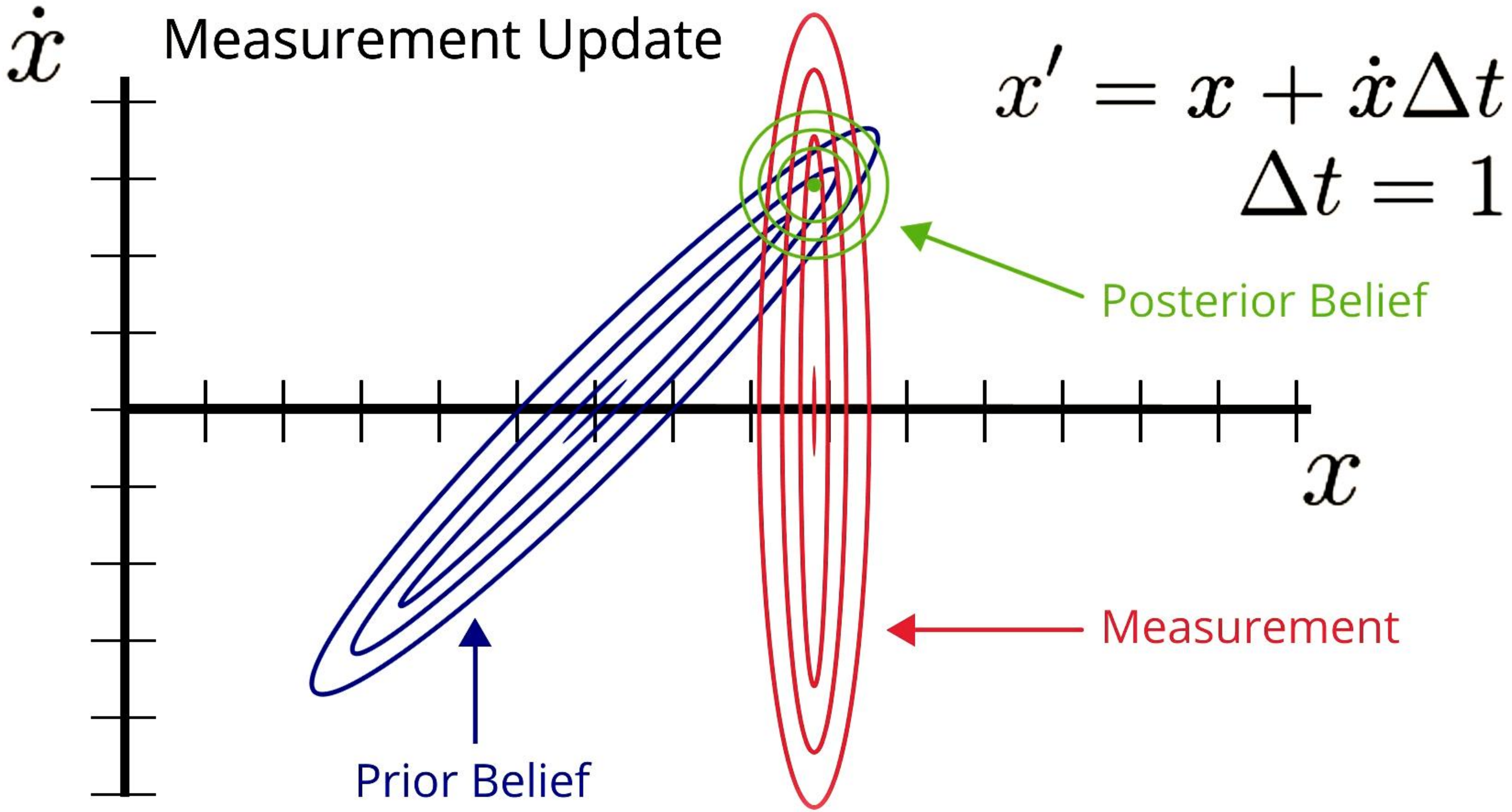


$$x' = x + \dot{x}\Delta t$$
$$\Delta t = 1$$

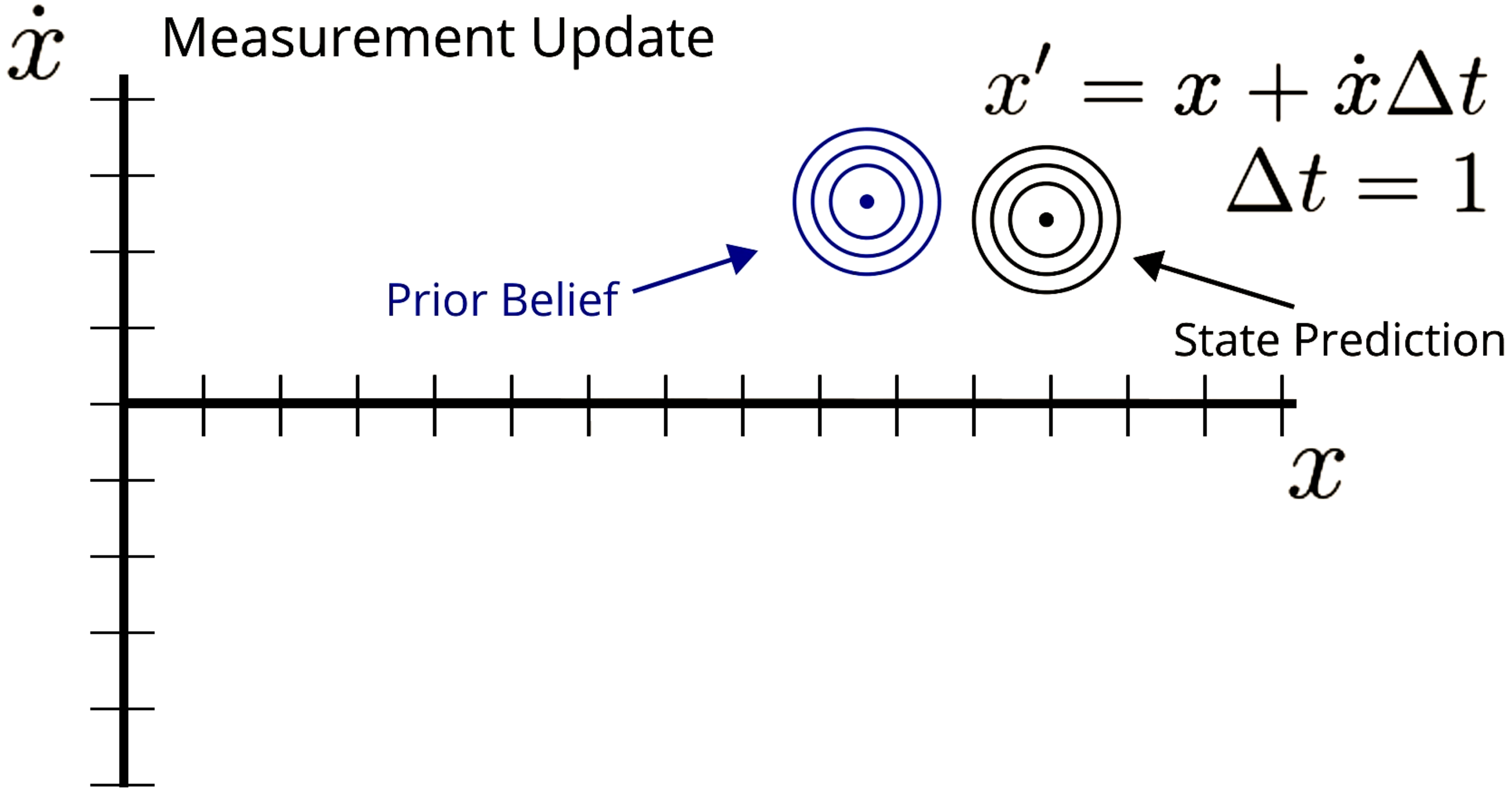
Multidimensional Kalman Filters – Measurement Update



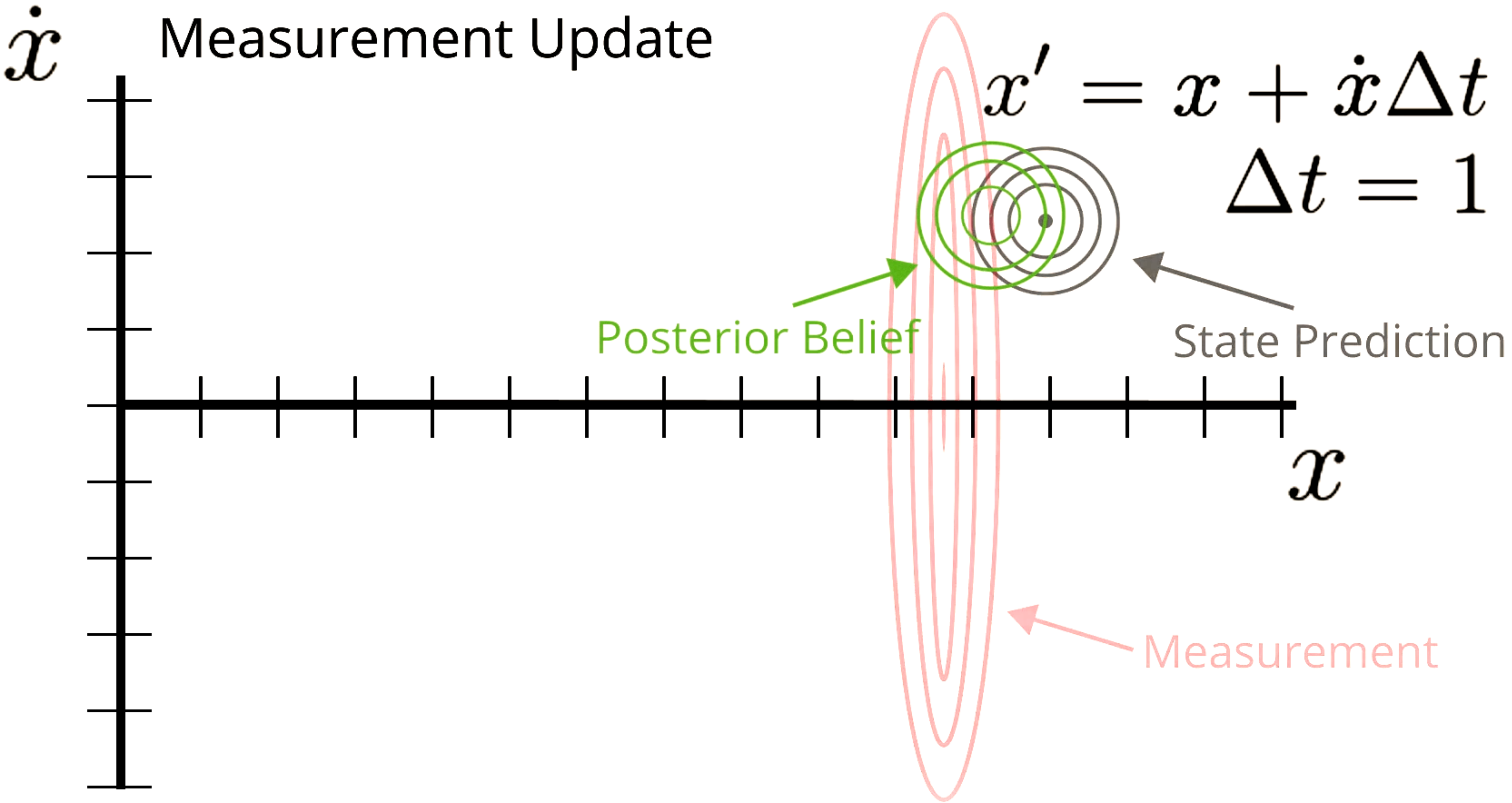
Multidimensional Kalman Filters – Measurement Update



Multidimensional Kalman Filters – State Prediction

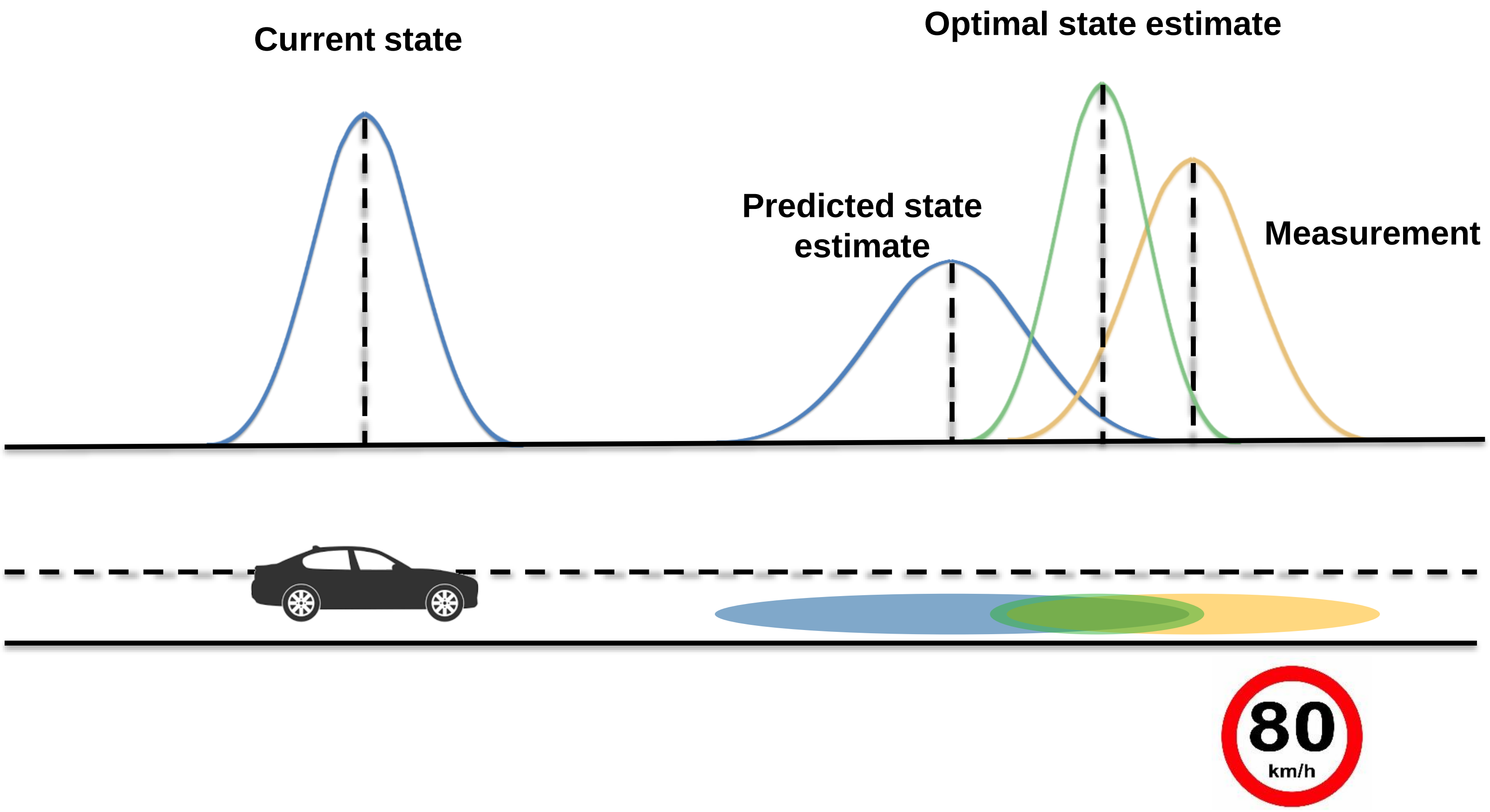


Multidimensional Kalman Filters – Measurement Update



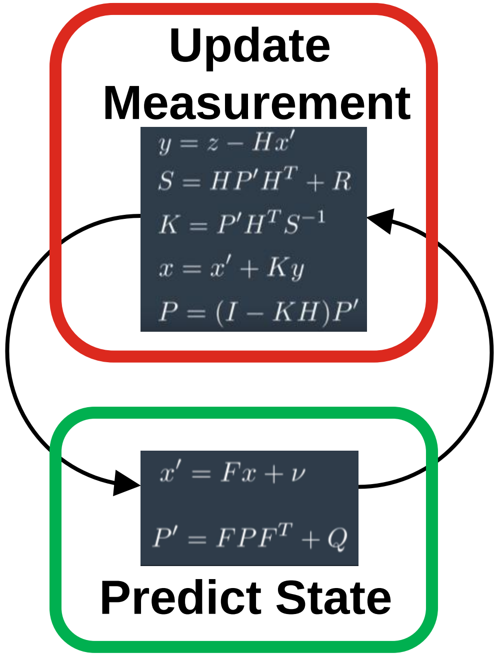
Kalman filter for localization

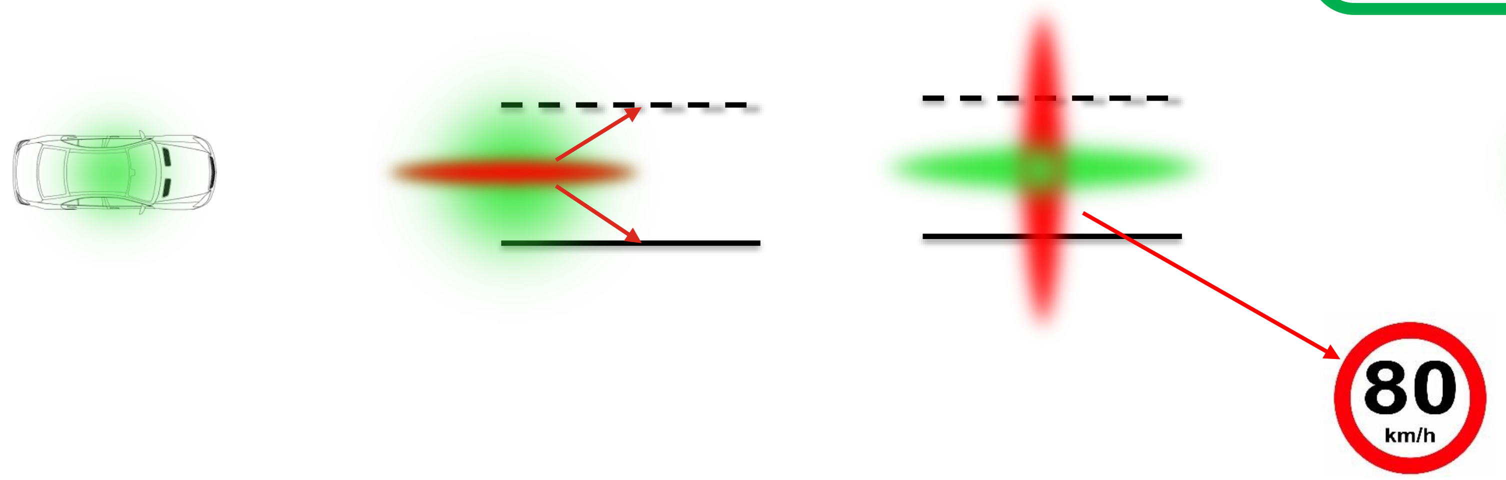
Kalman Filter for Localization



Map Matching Localization based on Kalman Filter

- **Lane measurement update**
  - Lateral correction
- **Traffic sign measurement update**
  - Longitudinal correction





THANK YOU
FOR YOUR ATTENTION